



Colorizing the Prokudin-Gorskii Photo Collection

Automatic Image Alignment and Color Channel Registration

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Background

Sergei Mikhailovich Prokudin-Gorskii (1863-1944) was a pioneering photographer who envisioned color photography and multi-media classrooms long before it was invented. Between 1907 and 1915, he traveled across the Russian Empire capturing thousands of images using a clever technique of recording three exposures of every scene onto glass plates using red, green, and blue filters.

The goal of this project is to automatically align these color channel images to produce a single RGB color photograph with minimal visual artifacts. This requires implementing efficient alignment algorithms that can handle both small JPEG images and large high-resolution TIFF scans.

Approach

Algorithm Overview

The core alignment pipeline consists of:

- **Channel extraction:** Splitting the stacked BGR image into separate channels
- **Alignment search:** Finding optimal (x, y) displacement using exhaustive or pyramid search
- **Quality check:** Automatic fallback to edge-based features if NCC score < 0.7
- **Post-processing:** Border cropping and automatic enhancements

Single-Scale Alignment

For smaller JPEG images, exhaustive search over a [-50, 50] pixel window is computationally feasible.
Single-scale alignment is applied full resolution, comparing the reference and target images over a predefined search range. Every possible shift within this range is evaluated to find the best match, using a similarity metric like normalized

cross-correlation or euclidean distance. While this approach works for small images, it becomes slow for large images or when large misalignments need to be corrected, since all possible translations must be exhaustively checked.

Multi-Scale Pyramid Alignment

For high-resolution TIFF images, a coarse-to-fine pyramid approach dramatically reduces computation time:

- Recursively downsample images by factor of 2
- Align at coarsest scale with full [-50, 50] search window
- Scale up displacement by 2× at each level
- Refine with small +3/-3 pixel search at each finer level

Multi-Scale Pyramid Alignment speeds up image alignment by progressively searching at multiple resolutions. The process begins by creating smaller, downsampled versions of the images, which reduces the number of pixels and makes a broad search for large shifts much faster. Once a rough alignment is found at the coarsest level, the estimated shift is scaled up to match the higher-resolution images and used as a starting point for a finer, local search. This recursive approach repeats across levels, gradually refining the alignment while only exploring small neighborhoods at each stage. By combining coarse, low-resolution estimates with fine, high-resolution adjustments, the method efficiently handles large misalignments.

Design Decisions and Optimizations

NCC vs L2 Norm

Normalized Cross-Correlation (NCC) performed better than L2 norm (Euclidean distance). NCC compares the patterns of pixel values rather than their absolute brightness. This makes it insensitive to differences in brightness and contrast, which is better since the R, G, and B channels may have different intensity levels due to different spectral sensitivity of the original glass plates.

Key Parameters

- **max_shift (50 pixels):** Large enough to handle typical misalignments in high-res scans
- **max_levels (6):** Balances pyramid depth with computational cost
- **min_size (50 pixels):** Prevents over-downsampling that would lose structural information
- **refine_range (3 pixels):** Small refinement window sufficient after coarse alignment

Border Cropping Strategy

Two stages of cropping improve alignment quality:

1. **During alignment:** Cropping 10% from borders during similarity scoring prevents edge artifacts from `np.roll` from affecting the metric
2. **Final output:** Final 10% crop removes remaining artifacts at the edges

Green as Reference Channel

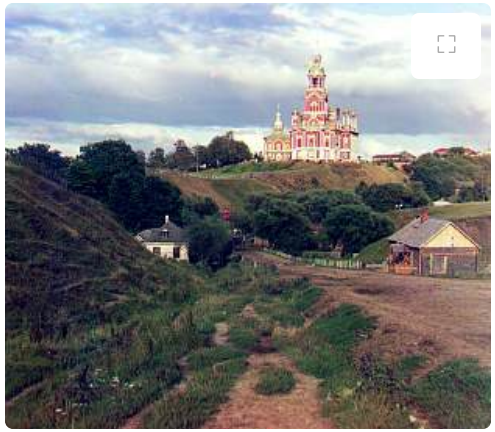
Instead of aligning to the Blue channel, this implementation uses Green as the reference. The green channel typically has the best/highest signal-to-noise ratio and intermediate sensitivity in the visible spectrum. Blue channels tend to be noisier due to atmospheric scattering. This choice proved especially effective for the Emir image, where blue channel overexposure caused poor alignment when aligned with blue channel as reference.

Automatic Edge-Based Fallback

When initial NCC-based alignment produces a score below 0.7, the algorithm automatically retries using Sobel edge features. This hybrid approach handles cases where brightness differences between channels are too severe for intensity-based alignment, while avoiding the computational overhead of edge detection for images that are easier to align. Edge features align based on structural content rather than absolute intensities, making them robust to exposure variations.

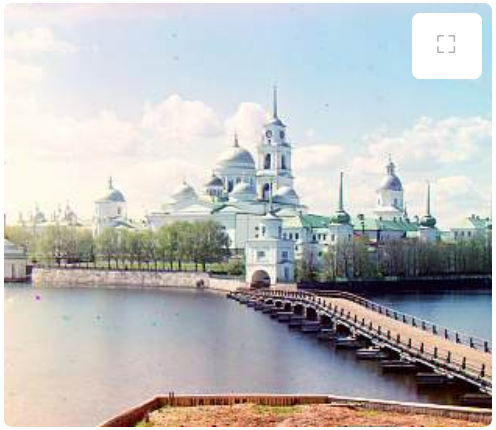
Single-Scale Alignment Results

Results on low-resolution JPEG images using single-scale exhaustive search with NCC metric:



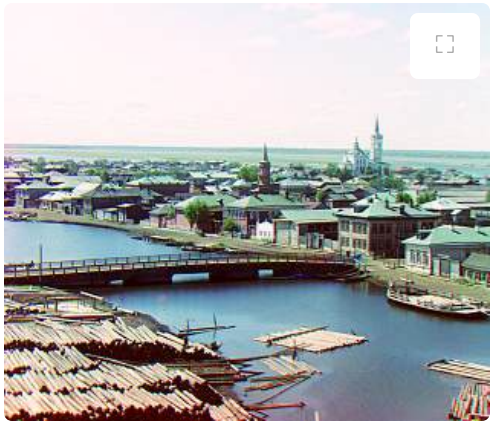
Cathedral

`R: (1, 7) • G: (0, 0) • B: (-2, -5)`



Monastery

`R: (1, 6) • G: (0, 0) • B: (-2, 3)`



Tobolsk

`R: (1, 4) • G: (0, 0) • B: (-3, -3)`

Image	Red Offset (x, y)	Green Offset (x, y)	Blue Offset (x, y)
cathedral.jpg	(1, 7)	(0, 0)	(-2, -5)
monastery.jpg	(1, 6)	(0, 0)	(-2, 3)
tobolsk.jpg	(1, 4)	(0, 0)	(-3, -3)

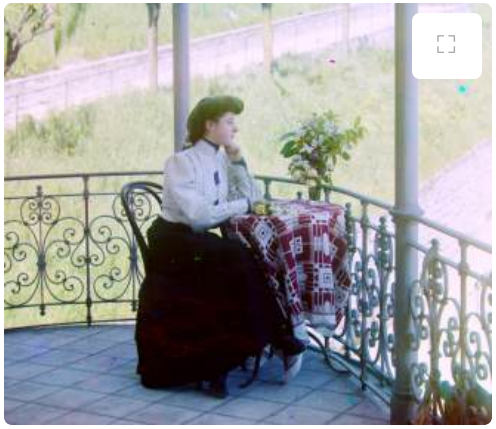
Multi-Scale Pyramid Alignment Results

Results on all images using pyramid alignment with NCC metric and automatic enhancements:



Emir

R: (17, -3152) • G: (0, 0) • B: (-24, -49)



Italil

R: (15, -3192) • G: (0, 0) • B: (-21, -3269)



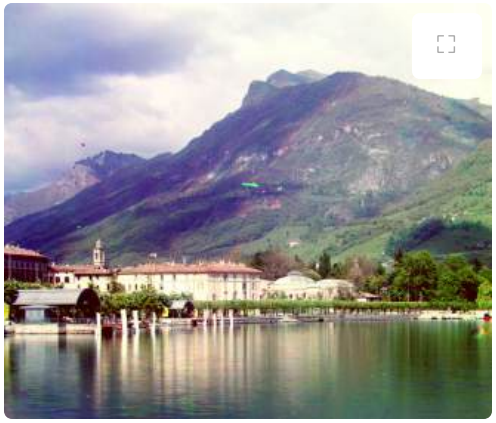
Church

R: (-8, -3169) • G: (0, 0) • B: (-4, -3227)



Three Generations

R: (-3, -3151) • G: (0, 0) • B: (-14, -53)



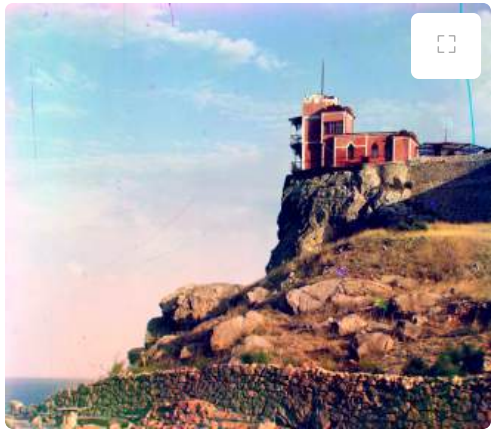
Lugano

R: (-13, -3192) • G: (0, 0) • B: (16, -41)



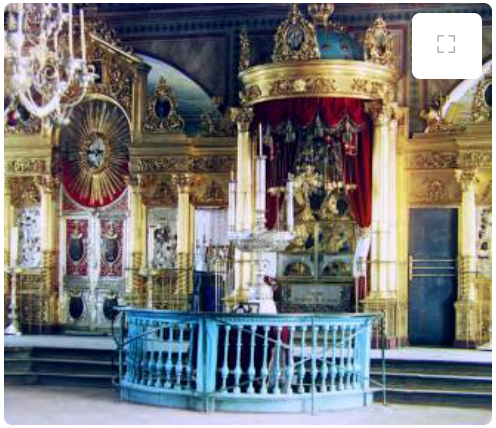
Melons

R: (4, -3145) • G: (0, 0) • B: (-10, -82)



Lastochikino

R: (-7, -3163) • G: (0, 0) • B: (2, -3238)



Icon

R: (5, -3196) • G: (0, 0) • B: (-17, -41)



Siren

R: (-18, -3203) • G: (0, 0) • B: (6, -49)



Self Portrait

R: (8, -3153) • G: (0, 0) • B: (-29, -79)



Harvesters

R: (-3, -3153) • G: (0, 0) • B: (-17, -60)

Image	Red Offset (x, y)	Green Offset (x, y)	Blue Offset (x, y)
cathedral.jpg	(1, 7)	(0, 0)	(-2, -5)
church.tif	(-8, -3169)	(0, 0)	(-4, -3227)
emir.tif	(17, -3152)	(0, 0)	(-24, -49)
harvesters.tif	(-3, -3153)	(0, 0)	(-17, -60)
icon.tif	(5, -3196)	(0, 0)	(-17, -41)
italil.tif	(15, -3192)	(0, 0)	(-21, -3269)
lastochikino.tif	(-7, -3163)	(0, 0)	(2, -3238)
lugano.tif	(-13, -3192)	(0, 0)	(16, -41)
melons.tif	(4, -3145)	(0, 0)	(-10, -82)
monastery.jpg	(1, 6)	(0, 0)	(-2, 3)
monument.tif	(24, -3183)	(0, 0)	(-27, -3268)
railtrack.tif	(-5, -3195)	(0, 0)	(-5, -52)
self_portrait.tif	(8, -3153)	(0, 0)	(-29, -79)
siren.tif	(-18, -3203)	(0, 0)	(6, -49)
three_generations.tif	(-3, -3151)	(0, 0)	(-14, -53)

Image	Red Offset (x, y)	Green Offset (x, y)	Blue Offset (x, y)
tobolsk.jpg	(1, 4)	(0, 0)	(-3, -3)
woods.tif	(37, -3176)	(0, 0)	(36, -3274)

Additional Examples from the Collection

Three additional images selected from the Prokudin-Gorskii collection:



Monument

R: (24, -3183) • G: (0, 0) • B: (-27, -3268)



Woods

R: (37, -3176) • G: (0, 0) • B: (36, -3274)



Railtrack

R: (-5, -3195) • G: (0, 0) • B: (-5, -52)

Monument	(37, -3176)	(0, 0)	(-27, -3268)
Woods	(37, -3176)	(0, 0)	(36, -3274)
Railtrack	(-5, -3195)	(0, 0)	(-5, -52)

Challenges

Church Image - Green Shadow Artifact Resolution

Initially, the Church image exhibited visible green shadow artifacts around buildings. This was caused by insufficient border cropping (5%) that failed to remove the misaligned edge regions created by the circular shift in `np.roll`.

Solution: Increasing the border crop percentage from 5% to 10% during both alignment scoring and final output eliminated these artifacts. The comparison below demonstrates the improvement.



Before: 5% border crop - green shadows visible



After: 10% border crop - artifacts eliminated

Bells and Whistles

1. Edge-Based Alignment Features

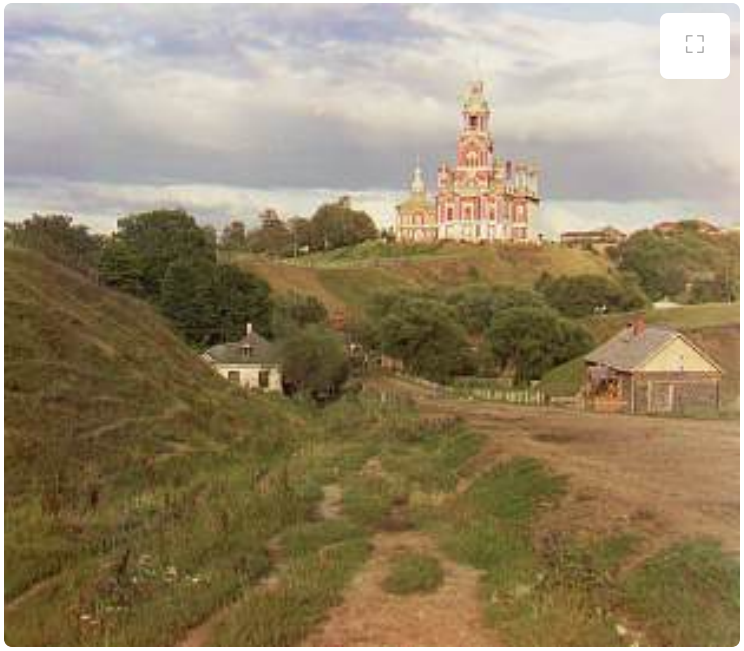
Edge-based alignment leverages the structural features of an image, such as edges and contours, rather than raw pixel intensities to achieve accurate registration. This approach is particularly useful when the color channels differ significantly in brightness or contrast, making standard intensity-based alignment unreliable. By focusing on gradients and edges, the method ensures that key shapes and patterns in the image are properly aligned.

Implemented automatic fallback to edge-based alignment when NCC score indicates poor alignment (score < 0.7), using Sobel gradient magnitude to align based on structural features rather than pixel intensities. The Sobel operator identifies regions where the brightness changes suddenly, indicating edges. This implementation first attempts standard NCC alignment, then checks the alignment quality. If the score is below threshold, it automatically retries using edge features.

2. Automatic Contrast Enhancement

Contrast refers to the difference in brightness between the light and dark areas of an image. Higher contrast makes details more distinguishable, while low contrast can make an image appear flat or washed out. Contrast enhancement techniques are used to improve the visibility of features by stretching or adjusting the intensity range of the image. This helps bring out subtle details, making dim or low-contrast images clearer and visually more informative.

After clipping the extreme 1% of values in each color channel to remove outliers, each color channel is independently normalized to utilize the full [0, 1] dynamic range.



Before: No enhancement



After: Contrast enhancement

View All Images –



Emir - Before



Emir - After



Italil - Before



Italil - After





Monastery - Before

Monastery - After



Church - Before



Church - After



Three Generations - Before



Three Generations - After





Lugano - Before

Lugano - After



Melons - Before



Melons - After



Lastochikino - Before



Lastochikino - After





Tobolsk - Before

Tobolsk - After



Icon - Before



Icon - After



Cathedral - Before



Cathedral - After





Monument - Before

Monument - After



Siren - Before

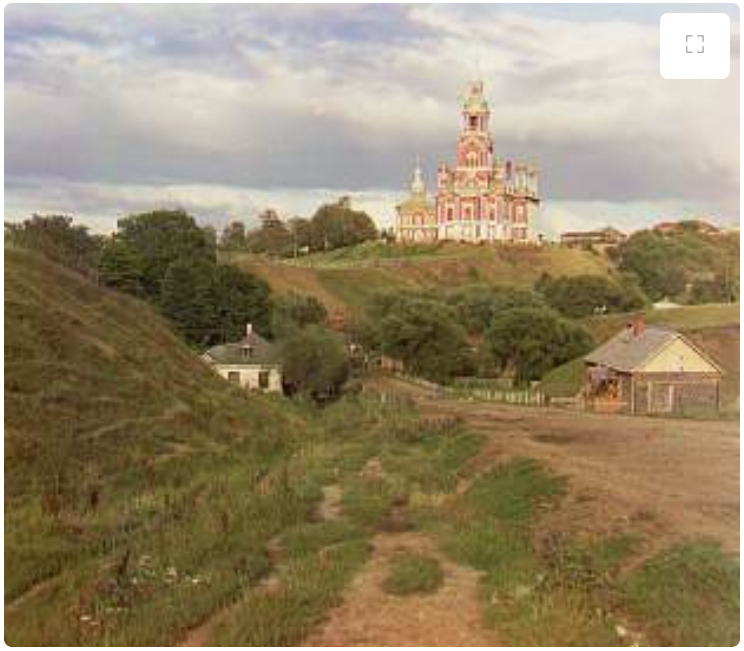


Siren - After

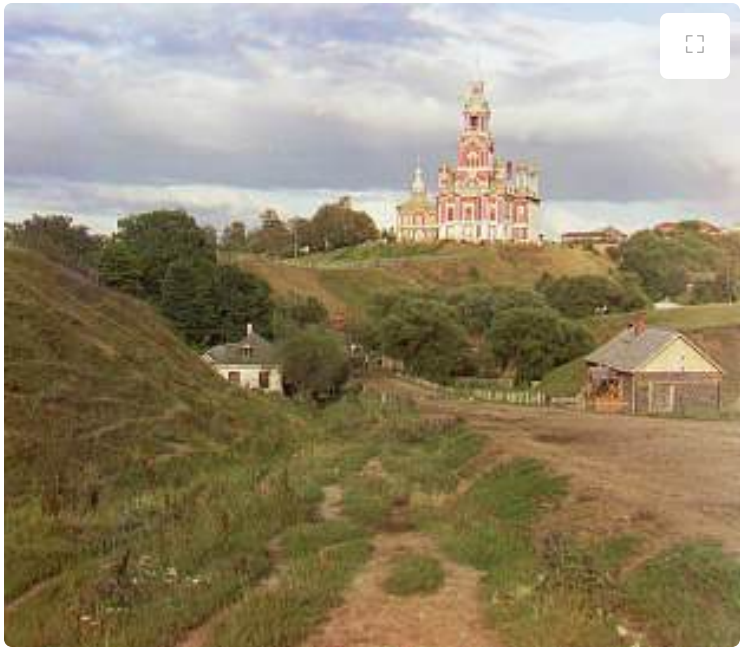
3. Automatic White Balance

White balance is the process of adjusting the colors in an image so that objects that appear white in real life are rendered as true white in the photo. This corrects for color casts caused by different lighting conditions, such as sunlight. White balance enhancement ensures that all colors in the image appear natural and consistent, making the overall scene look more realistic and visually accurate.

Implemented the White World algorithm, which assumes the brightest 5% of pixels should be white. This effectively removes color casts and normalizes illumination across channels.



Before: No enhancement



After: White balanced

View All Images



Cathedral - Before



Cathedral - After



Monastery - Before



Monastery - After





Tobolsk - Before

Tobolsk - After



Emir - Before



Emir - After



Church - Before



Church - After





Three Generations - Before

Three Generations - After



Lugano - Before



Lugano - After



Melons - Before



Melons - After





Lastochikino - Before

Lastochikino - After



Icon - Before



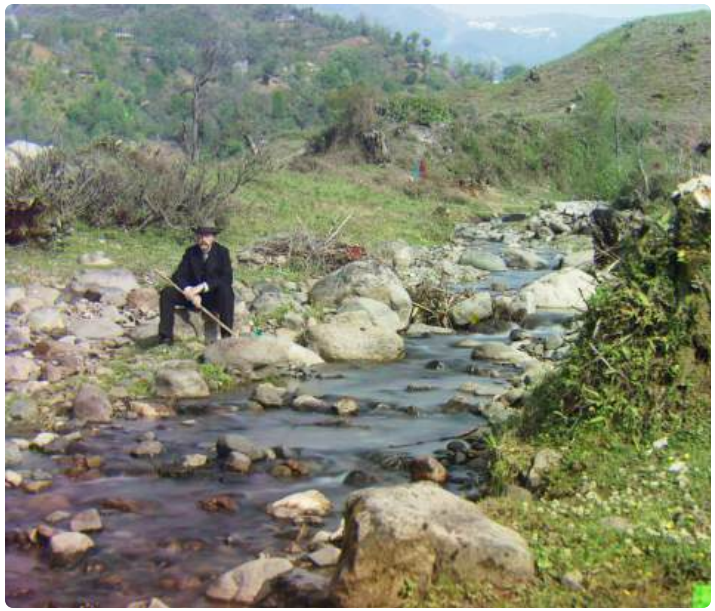
Icon - After



Siren - Before



Siren - After





Self Portrait - Before

Self Portrait - After



Harvesters - Before



Harvesters - After

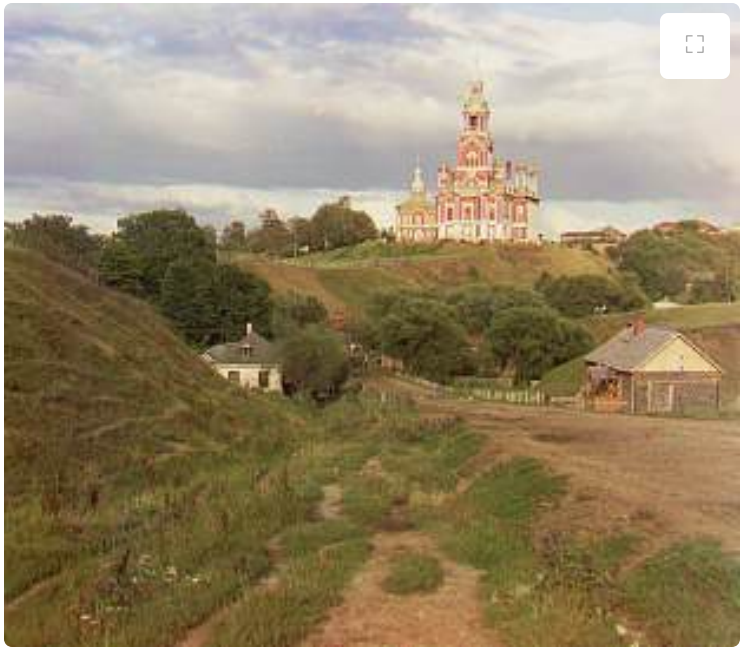




4. Adaptive Color Correction

Color correction is the process of adjusting the colors in an image to achieve a more natural and visually consistent appearance. Unlike white balance, which primarily removes color casts to ensure neutral whites, color correction focuses on balancing the overall color distribution across all channels. It compensates for differences in lighting, sensor characteristics, or historical capture methods, such as the varying RGB filters used in early photography, so that colors appear true to life and harmonized across the image.

Applied statistical color normalization that equalizes the mean and standard deviation across all three color channels. This helps compensate for differences in the original RGB filters used by Prokudin-Gorskii versus modern color spaces.



Before: No enhancement



After: Color corrected

View All Images –



Cathedral - Before



Cathedral - After



Monastery - Before



Monastery - After





Tobolsk - Before

Tobolsk - After



Emir - Before



Emir - After



Church - Before



Church - After





Three Generations - Before

Three Generations - After



Lugano - Before



Lugano - After



Melons - Before



Melons - After





Lastochikino - Before

Lastochikino - After



Icon - Before



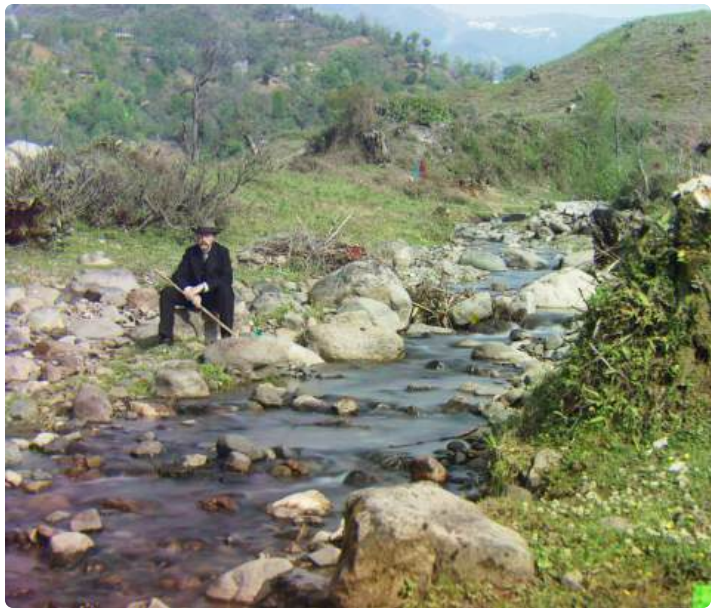
Icon - After



Siren - Before



Siren - After





Self Portrait - Before

Self Portrait - After



Harvesters - Before



Harvesters - After

Conclusion

This project successfully demonstrates automatic colorization of the Prokudin-Gorskii collection through multi-scale pyramid alignment and single-scale exhaustive search. The pyramid approach provides a massive speedup over naive exhaustive search while maintaining accuracy. All images from the provided dataset were successfully aligned with no failures.